Comparing the above equations, we get

$$L = \mu_{\circ} n^2 Al \qquad (4.10)$$

From the above equation, it is clear that inductance depends on the geometry of the solenoid (turn density n, cross-sectional area A, length l) and the medium present inside the solenoid. If the solenoid is filled with a dielectric medium of relative permeability μ_r, then

$$L = \mu n^2 Al \ \ or \ \ \ L = \mu_{\circ}\mu_r n^2 Al$$

Energy stored in an inductor

Whenever a current is established in the circuit, the inductance opposes the growth of the current. In order to establish a current in the circuit, work is done against this opposition by some external agency. This work done is stored as magnetic potential energy.

Let us assume that electrical resistance of the inductor is negligible and inductor effect alone is considered. The induced emf ε at any instant t is

$$\varepsilon = -L\frac{di}{dt}$$

Let dW be work done in moving a charge dq in a time dt against the opposition, then

$$dW = -\varepsilon dq$$
$$= -\varepsilon i dt \qquad \because dq = idt$$

Substituting for ε from equation (4.9),

$$= -\left(-L\frac{di}{dt}\right) i dt$$
$$dW = Lidi$$

Total work done in establishing the current i is

$$W = \int dW = \int_0^i Lidi = L\left[\frac{i^2}{2}\right]_o^i$$
$$W = \frac{1}{2}Li^2$$

This work done is stored as magnetic potential energy.

$$\therefore U_B = \frac{1}{2}Li^2 \qquad (4.11)$$

The energy density is the energy stored per unit volume of the space and is given by

$$u_B = \frac{U_B}{Al} \qquad \because \text{Volume of the solenoid} = Al$$
$$u_B = \frac{Li^2}{2Al} = \frac{\left(\mu_0 n^2 Al\right) i^2}{2Al} \qquad \because L = \mu_0 n^2 Al$$
$$= \frac{\mu_0 n^2 i^2}{2}$$
$$u_B = \frac{B^2}{2\mu_0} \qquad \because B = \mu_0 ni$$

EXAMPLE 4.10

A solenoid of 500 turns is wound on an iron core of relative permeability 800. The length and radius of the solenoid are 40 cm and 3 cm respectively. Calculate the average emf induced in the solenoid if the current in it changes from 0 to 3 A in 0.4 second.

Solution

N = 500 turns; $\mu_r = 800$;
l = 40 cm = 0.4 m; r = 3 cm = 0.03 m;
di = 3 – 0 = 3 A; dt = 0.4 s

Self inductance,

$$L = \mu n^2 Al \quad \left(\because \mu = \mu_{\circ}\mu_r; A = \pi r^2; n = \frac{N}{l}\right)$$
$$= \frac{\mu_{\circ}\mu_r N^2 \pi r^2}{l}$$
$$= \frac{4\times 3.14\times 10^{-7}\times 800\times 500^2\times 3.14\times\left(3\times 10^{-2}\right)^2}{0.4}$$
$$L = 1.77 \text{ H}$$

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